SYNTHETIC PROFESSIONAL-SERVICE SPECIMEN

Payment succeeded. Access failed.

Modernisation should not begin with a rewrite. AI should not begin with an agent. Both should begin with evidence.

EXECUTIVE CONCLUSION

The payment path was healthy. Fulfilment and access-state reconciliation were the risk.

SEVERITY	Critical
CONFIDENCE	High for selected workflow; wider exposure unknown.
X-RAY SPECIMEN	$2,250

BOUNDARY

Synthetic professional-service specimen—not a client case study or production monitoring product.

- Synthetic demonstration based on the completed Rescue Lab incident.
- Not a real client case, testimonial or production incident report.
- Not live monitoring, legal advice, compliance assurance or financial assurance.
- Commercial scope and pricing are illustrative specimen values, not a quotation or guarantee.
- Reusable proprietary assessment internals are intentionally redacted.

FULL X-RAY SPECIMEN

Evidence-backed decision document.

A Executive Risk Brief

INCIDENT

Payment succeeded. Access failed.

WORKFLOW

Payment -> webhook -> queue -> subscription -> entitlement -> customer access

CONSEQUENCE

A synthetic paid subscription remained locked beyond its entitlement SLA.

DIVERGENCE

Queue/job provisioning failed three bounded attempts after payment and idempotent event processing succeeded.

DECISION

Contain paid-but-locked exposure and approve a controlled X-Ray before rewrite or automation.

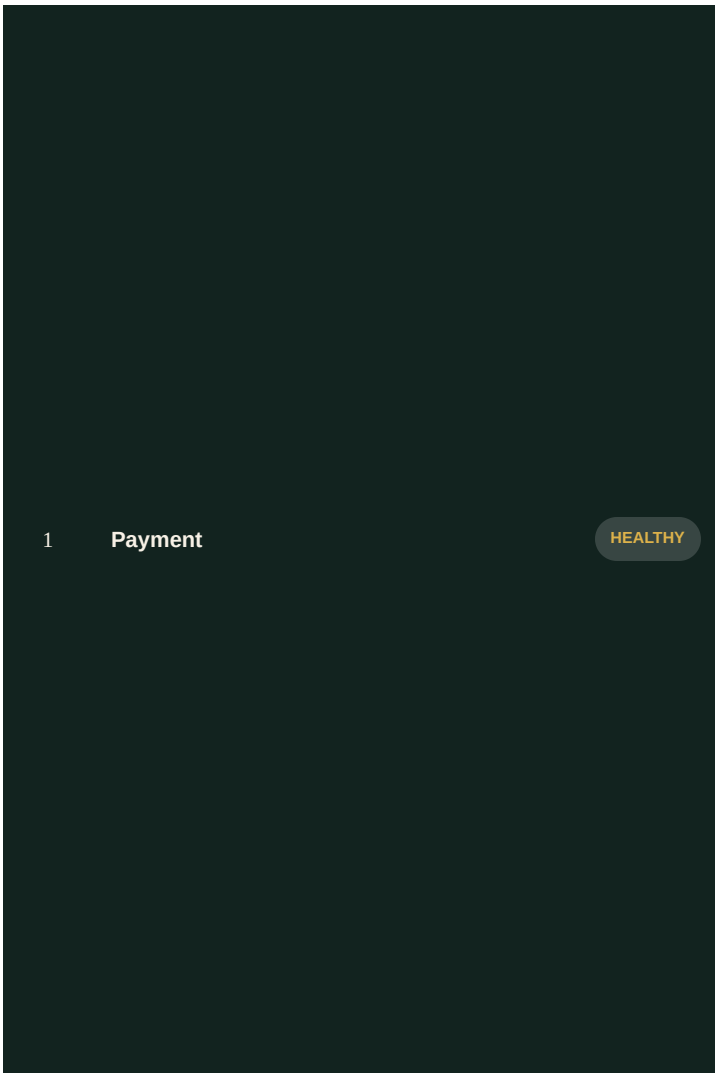
UNKNOWNS

A real client X-Ray must confirm code paths, data quality, ownership, monitoring and release constraints.

Confirmed impact: One synthetic account, USD 4,800 monthly subscription amount exposed while locked.

Possible wider exposure: Similar workflows that collect payment before asynchronous fulfilment may share the same hidden rules.

B Critical-Path Map



STAGE EVIDENCE

Payment

EXPECTED

Payment succeeds and remains authoritative.

OBSERVED

Payment succeeded; USD 4,800 synthetic monthly amount.

RULE

Do not rewrite payment history to hide internal fulfilment gaps.

RISK

Incorrect rollback could corrupt the source-of-truth cash record.

EV-PAYMENT

STAGE EVIDENCE

Webhook

EXPECTED

At-least-once provider delivery is accepted safely.

OBSERVED

10 deliveries retained for one provider event.

RULE

Each provider event has one logical business effect.

2 **Webhook** HEALTHY

RISK

Unsafe consumers can duplicate subscription or entitlement effects.

EV-WEBHOOK

STAGE EVIDENCE

Event / processing layer

EXPECTED

One accepted event advances subscription once.

OBSERVED

1 business action recorded.

RULE

Repeated delivery must not create repeated business effects.

RISK

Hidden replay behaviour can amplify later recovery work.

EV-WEBHOOK

FIRST DIVERGENCE EVIDENCE

Queue / job

EXPECTED

Entitlement provisioning succeeds within bounded attempts.

OBSERVED

3 failed attempts; deterministic timeout evidence retained.

3 **Event / processing layer** HEALTHY

RULE

Async fulfilment must either complete, expose failure or hand off to safe recovery.

RISK

This is the first proven divergence in the incident path.

EV-QUEUE

STAGE EVIDENCE

Subscription

EXPECTED

Paid subscription and access state remain consistent.

OBSERVED

Subscription active while entitlement stayed locked.

RULE

Active paid status alone does not prove customer value was delivered.

RISK

Commercial state can look healthy while the customer is blocked.

EV-PAYMENT EV-ENTITLEMENT

FIRST DIVERGENCE EVIDENCE

Entitlement

EXPECTED

One active entitlement exists for the subscription/key.

4 **Queue / job** DIVERGED

OBSERVED

One entitlement existed but state was locked.

RULE

Workspace/subscription/key uniqueness prevents duplicate access grants.

RISK

Manual repair without guards can create duplicate or conflicting entitlements.

EV-ENTITLEMENT

FIRST DIVERGENCE EVIDENCE

Customer access

EXPECTED

Customer receives access within SLA.

OBSERVED

Access remained locked for 10 minutes.

RULE

Payment-to-access SLA must surface paid-but-locked state.

RISK

The business consequence is a paid customer receiving no value after payment.

EV-INCIDENT

5 **Subscription** WARNING

6 **Entitlement** DIVERGED

7 **Customer access** BLOCKED

C Expected vs Actual State

STAGE	EXPECTED	INCIDENT	RECOVERED
Payment	Succeeded payment remains authoritative	succeeded	succeeded
Webhook	Duplicate delivery retained without duplicate effect	10 deliveries / 1 event / 1 effect	No rewrite needed
Queue	Provisioning completes or exposes bounded failure	3 failures	Handed to controlled reconciliation
Subscription	Active paid subscription matches access	active subscription / locked access	active subscription / active access
Entitlement	One active entitlement	one locked entitlement	one active entitlement
Remaining risk	Systemic exposure known	unknown outside fixture	still unknown outside fixture

BEFORE RECOVERY

Paid + active subscription + locked entitlement

Customer value not delivered.

AFTER RECOVERY

Existing entitlement active once

Verified in this synthetic scenario.

REMAINING RISK

Wider exposure unknown

A real X-Ray checks adjacent workflows.

D Evidence Timeline

15:07:34

PAYMENT / PAYMENT

Synthetic payment succeeded.

`53c6028e-c818-46ed-8c3b-8779fd0090d9`

EV-PAYMENT

F-01

succeeded

15:17:34

WEBHOOK / WEBHOOK

10 signed deliveries retained for one event.

`evt_payment_succeeded_fragile`

EV-WEBHOOK

F-02

accepted

15:17:34

QUEUE / QUEUE

Provisioning attempt 1 failed.

`b3698195-02a7-4821-b7cc-d8398b88bd7c`

EV-QUEUE

F-02

synthetic_provider_timeout

15:17:34

QUEUE / QUEUE

Provisioning attempt 2 failed.

`69d1145d-3f3c-4e8e-88ae-df2ef2a4972b`

EV-QUEUE

F-02

synthetic_provider_timeout

15:17:34

QUEUE / QUEUE

Provisioning attempt 3 failed.

1192f70f-be64-4bb6-bf75-d8f7a9235a2b

EV-QUEUE

F-02

synthetic_provider_timeout

15:17:34

ENTITLEMENT / ENTITLEMENT

Entitlement existed but remained locked.

329

EV-ENTITLEMENT

F-01

locked

15:17:34

AUDIT / ACCESS

Paid-but-locked SLA incident opened.

53c6028e-c818-46ed-8c3b-8779fd0090d9

EV-INCIDENT

F-01

open

15:17:35

APPROVAL / RECOVERY

Approved action is bounded reconciliation of the existing entitlement.

53c6028e-c818-46ed-8c3b-8779fd0090d9

EV-RECOVERY

F-03

approved specimen decision

15:17:36

RECONCILIATION / RECOVERY

Recovery activates one existing entitlement.

53c6028e-c818-46ed-8c3b-8779fd0090d9

EV-RECOVERY

F-03

repaired / 1 effect

15:17:37

IDEMPOTENCY-CHECK / RECOVERY

Repeat recovery produces no_op and zero effects.

53c6028e-c818-46ed-8c3b-8779fd0090d9

EV-RECOVERY

F-03

no_op / 0 effects

15:17:38

VERIFICATION / ACCESS

Verified outcome remains synthetic and scoped to the specimen workflow.

53c6028e-c818-46ed-8c3b-8779fd0090d9

EV-STABILISED

F-05

verified in Lab

E Root-Cause Decision

PROXIMATE FAILURE

The entitlement provisioning job exhausted bounded synthetic timeouts.

UNDERLYING CONTROL GAP

The paid-access path needed evidence-backed detection and controlled reconciliation after async fulfilment failure.

WHY NOT EARLIER

Payment and subscription state looked healthy until payment-to-access evidence was reconciled against entitlement state and SLA.

CONFIDENCE

High for the synthetic workflow; unknown beyond it.

Competing explanations considered

Payment provider failure

Rejected for this fixture: payment is succeeded and authoritative.

EV-PAYMENT

Duplicate webhook created duplicate customer effects

Rejected for this fixture: one event and one business effect were recorded.

EV-WEBHOOK

Access failed because no subscription existed

Rejected for this fixture: subscription became active while entitlement stayed locked.

EV-ENTITLEMENT

F Hidden Business-Rule Register

BR-01 / HIGH

Successful payment does not independently prove access.

PURPOSE

Protect customer-value delivery after cash collection.

ENFORCEMENT

RevenueIntegrityAuditor / dashboard evidence

FAILURE CONSEQUENCE

Paid customer can remain locked without detection.

OWNER

Engineering + Revenue Ops

TEST/CONTROL

Feature test for paid-but-locked detection.

EV-PAYMENT

BR-02 / HIGH

Repeated provider delivery must not create duplicate business effects.

PURPOSE

Protect subscription and entitlement state from replay.

ENFORCEMENT

WebhookIngestor unique provider event and idempotent consumer

FAILURE CONSEQUENCE

Duplicate entitlements or repeated transitions.

OWNER

Platform Engineering

TEST/CONTROL

Duplicate delivery feature test.

EV-WEBHOOK

BR-03 / HIGH

Active paid status requires a consistent entitlement.

PURPOSE

Align commercial state with delivered value.

ENFORCEMENT

Auditor and entitlement constraints

FAILURE CONSEQUENCE

Support tickets, churn risk and unsafe manual fixes.

OWNER

Product Engineering

TEST/CONTROL

Expected-vs-actual reconciliation assertion.

EV-ENTITLEMENT

BR-04 / HIGH

Stale or invalid events must not overwrite newer state.

PURPOSE

Prevent provider noise from corrupting lifecycle state.

ENFORCEMENT

Signature verification and guarded transitions

FAILURE CONSEQUENCE

Access can be cancelled or mutated by unsafe replay.

OWNER

Platform Engineering

TEST/CONTROL

Out-of-order and invalid event tests.

EV-WEBHOOK

BR-05 / HIGH

BR-06 / SPECIMEN INFERENCE

Recovery requires audit, idempotency and verification.

PURPOSE

Make repair safe to run under pressure.

ENFORCEMENT

ReconciliationService transaction and immutable actions

FAILURE CONSEQUENCE

Manual repair can duplicate or hide the true incident.

OWNER

Engineering Lead

TEST/CONTROL

Repeat no-op recovery test.

EV-RECOVERY

Rollback must not corrupt authoritative payment records.

PURPOSE

Keep cash evidence separate from fulfilment repair.

ENFORCEMENT

Operations runbook

FAILURE CONSEQUENCE

Financial truth can be obscured by internal state rewrites.

OWNER

Engineering + Finance owner

TEST/CONTROL

Rollback checklist and audit review.

EV-PAYMENT

G Prioritised Findings

Severity model: Customer/revenue impact / Likelihood or exposure / Detectability / Recoverability / Evidence confidence

CRITICAL

CONFIRMED DEFECT

F-01. Paid customer locked after successful payment

One synthetic paid account received no access beyond SLA.

WORKFLOW

Payment-to-access workflow

RESPONSE

Contain and reconcile affected paid subscriptions before broad changes.

URGENCY

Immediate

ACCEPTANCE

Query finds zero paid active subscriptions with locked entitlements beyond SLA.

EV-PAYMENT EV-ENTITLEMENT EV-INCIDENT

HIGH

SYSTEMIC CONTROL GAP

F-02. Async fulfilment exhausted retries without closure

Retries were visible but did not complete access or trigger safe automatic closure.

WORKFLOW

Queue / entitlement job

RESPONSE

Add bounded failure handling, alerting and operator handoff.

URGENCY

24-48 hours

ACCEPTANCE

Exhausted jobs create actionable incident evidence.

EV-QUEUE

HIGH

CONFIRMED RISK CONTROLLED IN LAB

F-03. Manual repair needs repeat-safety proof

A duplicate repair path could create multiple entitlements or hide state divergence.

WORKFLOW

Reconciliation

RESPONSE

Use lock-protected reconciliation and run repeat no-op verification.

URGENCY

Before production repair

ACCEPTANCE

Second repair returns no_op and entitlement count remains one.

EV-RECOVERY

MEDIUM

UNVERIFIED RISK

F-04. Wider exposure is unknown outside the selected path

Other paid workflows may share the same hidden business rules.

WORKFLOW

Portfolio / similar workflows

RESPONSE

Scope a real X-Ray against one selected client workflow.

URGENCY

Before modernisation scope

ACCEPTANCE

Approved evidence confirms or rejects similar exposure.

EV-UNKNOWN

MEDIUM

IMPROVEMENT OPPORTUNITY

F-05. Evidence standard should precede rewrite or AI automation

Unsafe automation can accelerate an unresolved state problem.

WORKFLOW

Modernisation decision

RESPONSE

Modernise the critical path after evidence, tests and rollback are in place.

URGENCY

Before sprint planning

ACCEPTANCE

Critical-path map, tests and rollback rehearsal accepted.

EV-STABILISED

H Immediate Containment Plan

ENGINEERING OWNER

Identify paid-but-locked accounts

PRECONDITION

Database access to payment, subscription and entitlement state.

ACTION

Run scoped reconciliation query by workspace and SLA.

EVIDENCE CAPTURED

Query result and account list.

RISK

False positives if SLA rule is wrong.

REVERSAL

Read-only query can be discarded.

VALIDATION

Sample accounts checked against payment and access state.

LIMITATION

Does not prove wider workflows are healthy.

ENGINEERING LEAD

Freeze broad replay/retry actions

PRECONDITION

Incident commander agrees recovery boundary.

ACTION

Pause bulk queue retries and manual entitlement creation.

EVIDENCE CAPTURED

Decision note and queue state.

RISK

May delay unrelated work.

REVERSAL

Resume only after root cause and repair path are confirmed.

VALIDATION

No new duplicate entitlements appear.

LIMITATION

Does not fix the locked account.

APPLICATION ENGINEER

Run controlled recovery once

PRECONDITION

Payment, subscription and entitlement identifiers confirmed.

ACTION

Execute transactionally scoped reconciliation.

EVIDENCE CAPTURED

Before/after snapshot and action ID.

RISK

Incorrect scope could touch wrong workspace.

REVERSAL

Forward correction; do not rewrite payment history.

VALIDATION

Access active; entitlement count one.

LIMITATION

Only covers selected incident.

APPLICATION ENGINEER

Verify repeat no-op

PRECONDITION

First recovery completed and audited.

ACTION

Run the same reconciliation again.

EVIDENCE CAPTURED

No-op action ID and zero effects.

RISK

Hides defects if evidence not inspected.

REVERSAL

Stop and investigate if effects are non-zero.

VALIDATION

Second result is no_op; entitlement count remains one.

LIMITATION

Does not remove need for root repair.

Do not do yet

- Do not rewrite the billing system before evidence is mapped.
- Do not blindly replay every provider event or failed job.
- Do not create replacement entitlements without uniqueness and audit.
- Do not edit payment-provider history to make internal state appear consistent.
- Do not add an AI agent to a workflow whose business rules are still hidden.

I Recovery and Rollback Plan

APPROVED ACTION

Activate the existing locked entitlement through ReconciliationService.

PREREQUISITES

Confirm workspace, payment, subscription, entitlement and incident identifiers.

BOUNDARY

One workspace, one incident, existing entitlement only.

IDEMPOTENCY

Unique entitlement plus row locks; repeat action returns no_op and zero effects.

VERIFICATION

- payment remains succeeded
- subscription remains active
- entitlement active
- access restored
- repaired action recorded
- repeat no_op recorded

ROLLBACK TRIGGER

Unexpected workspace, entitlement count greater than one, repeat effect greater than zero, or mismatched payment/subscription evidence.

ROLLBACK ACTION

Stop further repair and apply forward correction after evidence review; do not rewrite payment history.

MONITORING

Watch new paid-but-locked incidents, failed provisioning attempts and duplicate-effect counts.

OWNER

Engineering lead with revenue owner approval.

J Verified Outcome

VERIFIED IN SYNTHETIC SCENARIO

Payment remains authoritative

EV-PAYMENT

VERIFIED IN SYNTHETIC SCENARIO

Subscription state is correct

EV-PAYMENT

VERIFIED BY COMMAND/TEST

Entitlement becomes active once

EV-RECOVERY

VERIFIED IN SYNTHETIC SCENARIO

Access is restored

EV-RECOVERY

VERIFIED BY IMMUTABLE ACTION MODEL/TEST

Recovery is audited

EV-RECOVERY

VERIFIED BY COMMAND/TEST

Repeat recovery is a no-op

EV-RECOVERY

NOT PRODUCTION PROVEN

Systemic risks remain visible

EV-UNKNOWN

K Priced 30-Day Rescue Roadmap

ILLUSTRATIVE CRITICAL SOFTWARE X-RAY

$2,250

ILLUSTRATIVE 30-DAY SPRINT

$9,500

X-Ray fee may be credited if the sprint is signed within 14 days. Illustrative specimen pricing-not a quotation or guarantee.

DAYS 1-5

Containment and reconciliation

$2,500

Affected paid-access state is bounded and repaired safely.

DELIVERABLES

paid-but-locked query; controlled recovery path; repeat no-op verification

ACCEPTANCE

Known affected records reconciled; no duplicate entitlements created.

BUYER RESPONSIBILITY

Provide approved access and name the workflow owner.

DAYS 6-12

Root repair and business-rule tests

$2,500

Hidden business rules are captured as executable controls.

DELIVERABLES

Business Rule Atlas; feature tests; state-transition guardrails

ACCEPTANCE

Tests fail on paid-but-locked and duplicate-effect regressions.

BUYER RESPONSIBILITY

Confirm rule owners and SLA thresholds.

DAYS 13-21

Observability, idempotency and failure handling

$2,500

Retries, inbox state and recovery signals become operationally visible.

DAYS 22-30

Controlled release, rollback rehearsal and handover

$2,000

The critical path can be changed without blind production risk.

DELIVERABLES

DELIVERABLES

correlation trail; failure handoff; idempotency evidence

ACCEPTANCE

A replayed provider event creates one business effect and visible evidence.

BUYER RESPONSIBILITY

Approve operational alert thresholds.

release checklist; rollback rehearsal; engineering handover

ACCEPTANCE

Release and rollback steps are rehearsed with evidence captured.

BUYER RESPONSIBILITY

Schedule release window and responsible approver.

Assumptions

- One selected critical workflow.
- Approved access to relevant code and sanitized evidence.
- No production credentials required for the first triage call.

Exclusions

- Full rewrite.
- Security/compliance certification.
- Financial audit.
- Broad codebase review outside selected path.
- AI copilot or autonomous agent build.

Scope-change conditions

- Multiple workflows.
- Missing evidence.
- Required infrastructure change.
- Security or compliance blockers.

L Real X-Ray Deliverables

Critical-path map

Evidence and state reconciliation

Business Rule Atlas sample

Prioritised risks

Immediate containment decision

Recovery and rollback plan

Baseline and verification criteria

Priced 30-day rescue roadmap

Executive readout

The X-Ray is not “a PDF.” It is an evidence-backed decision that reduces the risk of unsafe development, rewriting or automation.

M Evidence Register

EV-PAYMENT / OBSERVED

Payment confirmation

`payments`

pay_northstar_fragile / succeeded / USD 4,800

High

EV-WEBHOOK / OBSERVED

Signed webhook inbox

`webhook_events + webhook_deliveries`

evt_payment_succeeded_fragile / 10 deliveries / 1 event

High

EV-QUEUE / OBSERVED

Provisioning attempts

`provisioning_attempts`

3 attempts / 3 failures

High

EV-ENTITLEMENT / OBSERVED

Entitlement state

`entitlements`

Entitlement 329 / locked

High

EV-INCIDENT / OBSERVED

SLA incident

`incidents`

paid_but_locked / open / USD 4,800

High

EV-RECOVERY / VERIFIED

Recovery design

`ReconciliationService + PaidAccessRescueTest`

Locked entitlement -> active once; repeat recovery -> no_op / 0 effects

High

EV-STABILISED / OBSERVED

Stabilised replay

`lab_runs`

10 deliveries / 1 action / active access

High

EV-UNKNOWN / UNKNOWN

Client-specific wider exposure

`real X-Ray scope`

Requires approved client evidence; not inferred from this Lab

Explicit unknown